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Requirement Specification

Security Access Methodology for Flashing ECU over UDS

Designer	Lead	HOD	Stakeholder
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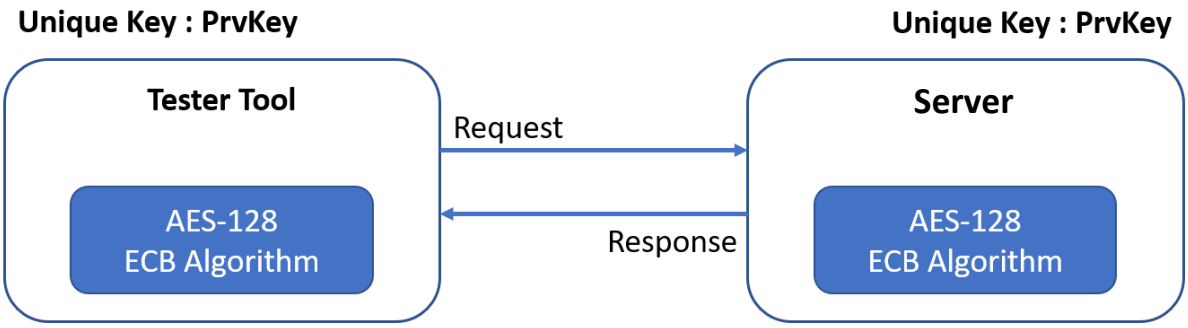
REVISION HISTORY

Sl. No	Revision No	Remarks	Date
1	1	Preliminary	25/10/2020
2	2	Added Example Implementation and Init Vector	31/12/2020
3	3	Changed the algo to AES 128 ECB	02/03/2021
4	4	Added the Flow chart of Protocol	26/03/2021
5	5	Made the document generic	09/02/2024

CHANGES MADE

Sl. No	Revision No	Remarks	Date
1	2	Init Vector updated in Algorithm	31/12/2020
1	2	Algo reverted back to AES128 ECB	02/03/2021

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Sequence of Steps for the Security Access:

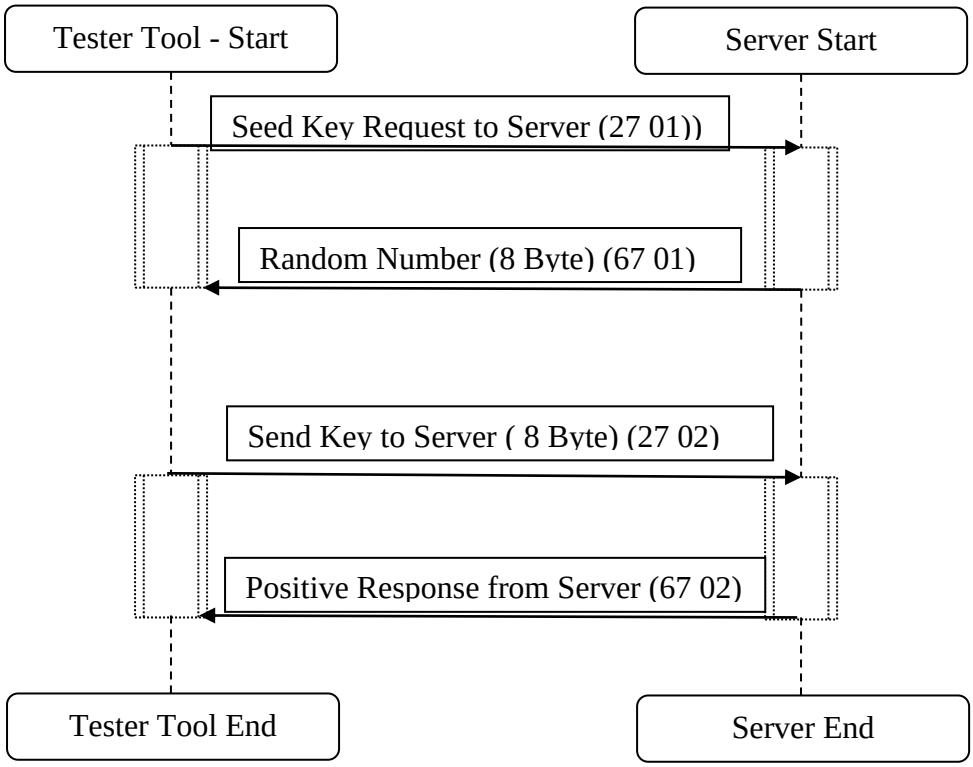
1. Tester Tool requests for the Seed Value to Server.
2. Server Responds to the Seed value request by sending the Seed Value.
   Seed value – 64-bit (0 – 7 Bytes of Random number)
3. Tester Tool receives the Seed Value and Encrypts the Seed with PrvKey using AES-128 ECB Algorithm, and generates the Encrypted Key
4. Tester Tool Transmits the 64bit (4 – 11th Byte) of Encrypted Key to Server
5. Server receives the 64bit of Encrypted Key,
6. Server Encrypts internally the combination of Random number (8 bytes) and Fixed Code (8 bytes) with PrvKey using AES-128 ECB Algorithm, and generated the Encrypted Key
7. Server Compares the 64bit of Encrypted Key generated internally with the Encrypted Key received from Tester in Step 4
8. If the Internally Generated Encrypted Key value is same as Received Encrypted Key value from Tester, the Server Authenticates the Tester tool, and releases the Security Level for Programming Server, and sends a Positive Response
   Else, it locks the Server and sends a Negative Response

Prerequisites:

a) AES-128 ECB Algorithm implemented on both Tester Tool and Server
b) PrvKey, is a Unique key(128-bit) that needs to be stored in both Tester Tool and Server
c) Fixed Code – (8Bytes – 64Bits) to be stored in both Tester Tool and Server
d) Negative Response Code: 0x33, 0x35 will be sent.

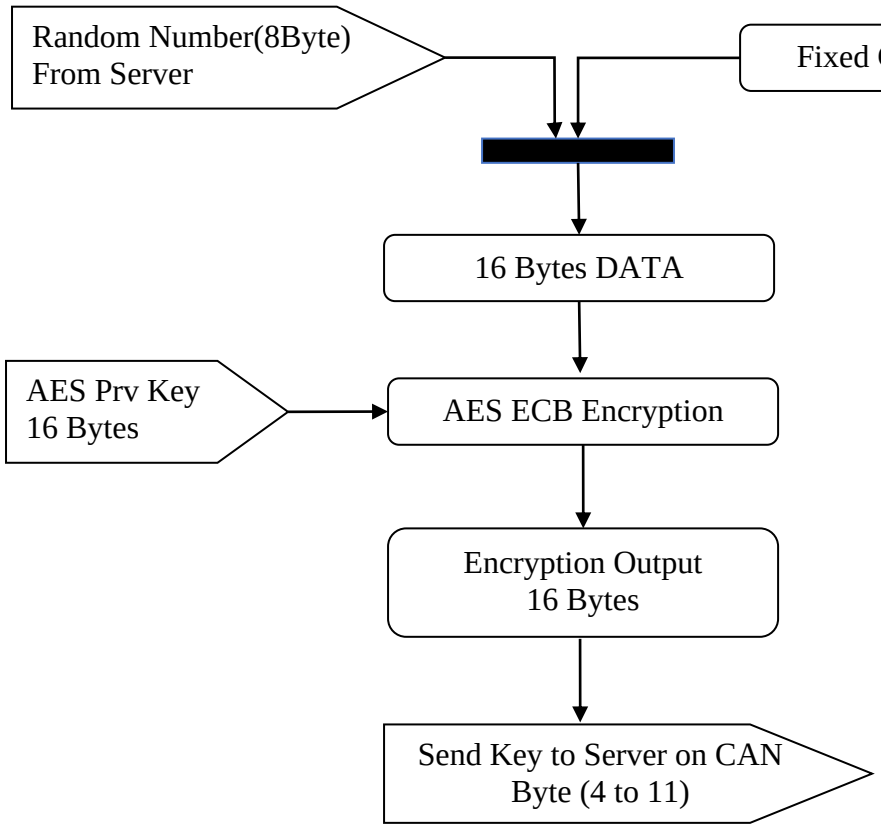
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Protocol of Transmission (Security Access Service)



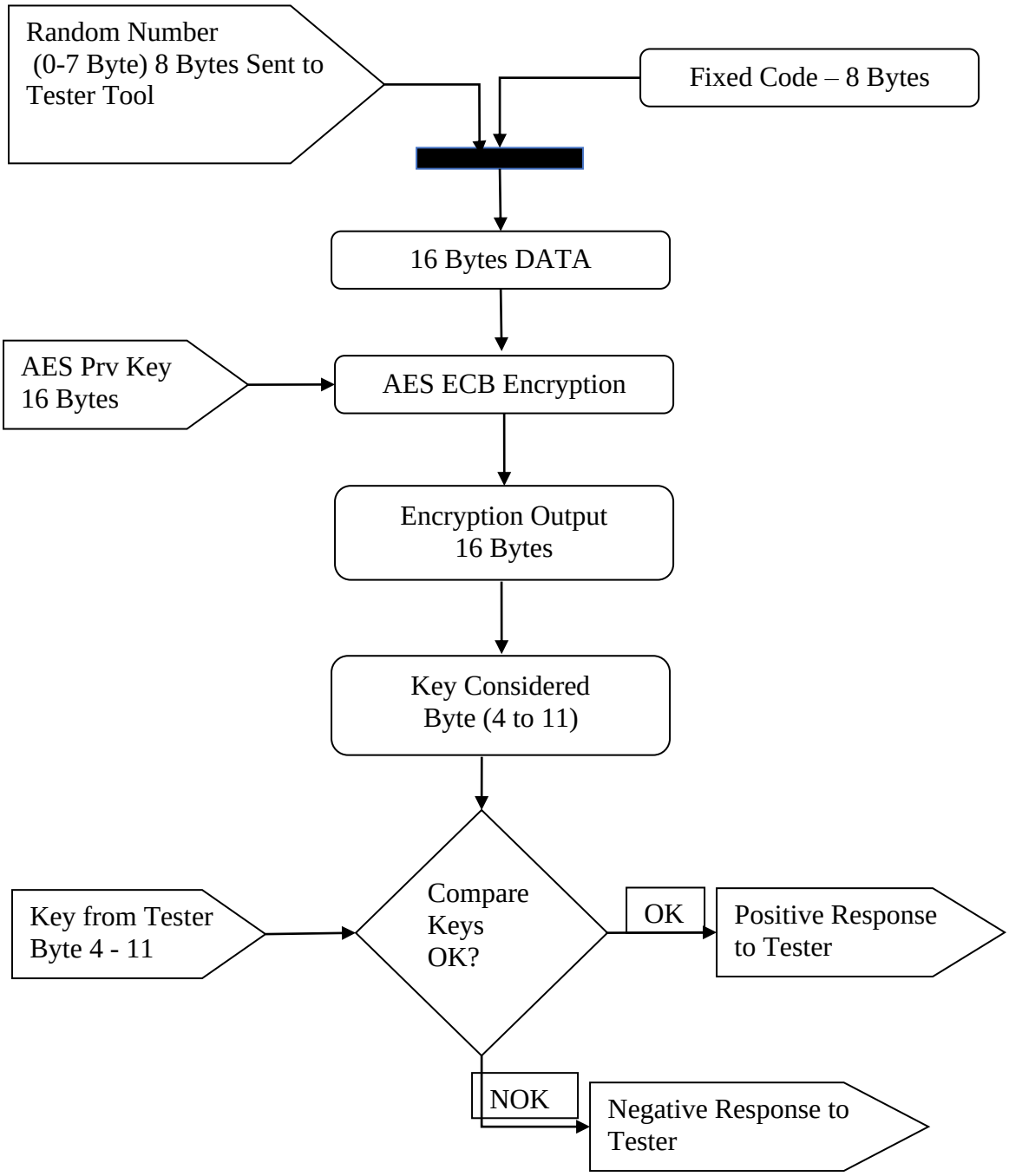
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Authentication Flow at Tester Tool



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Authentication Flow at Server



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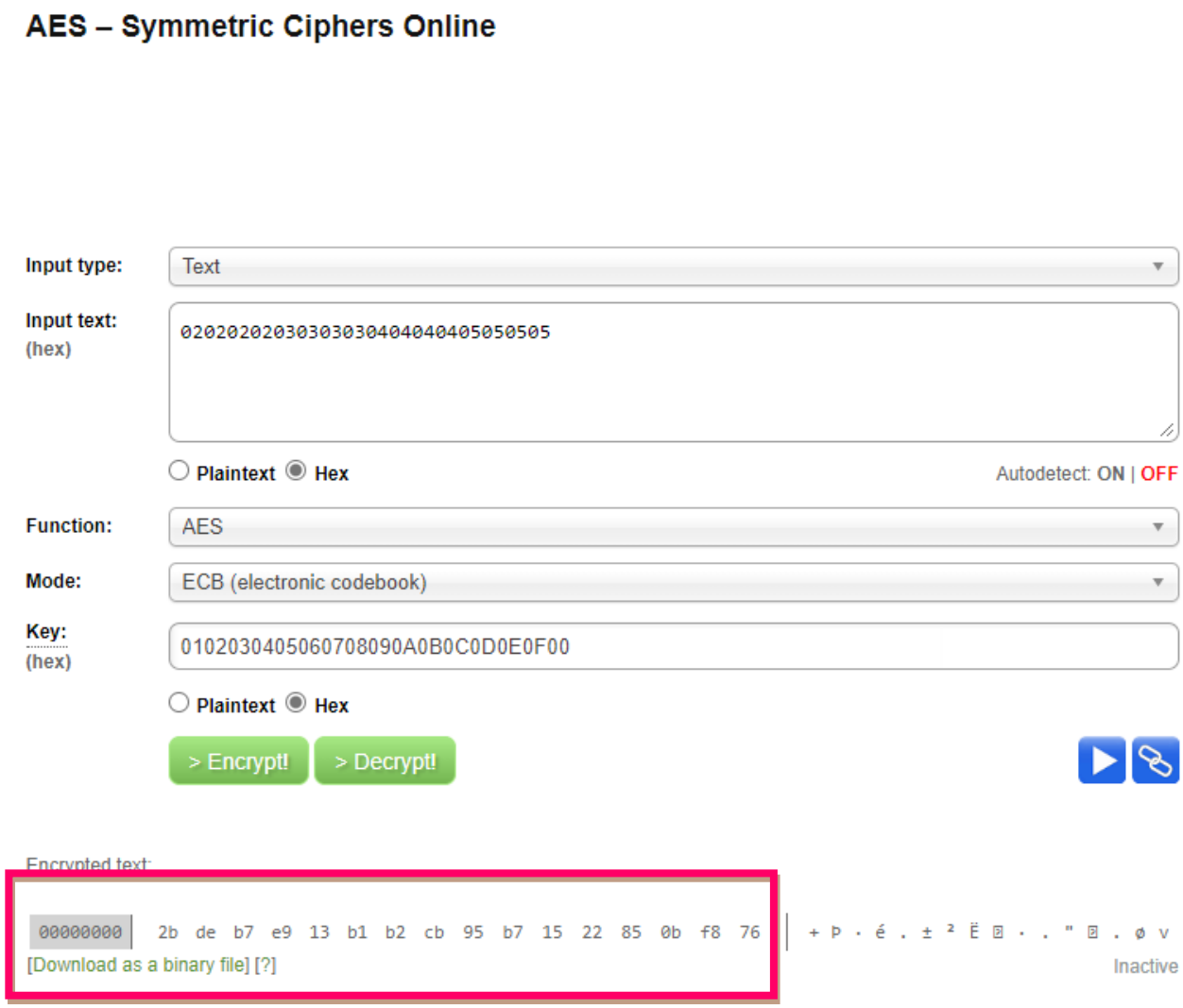
Example Implementation:

Please use below link to AES 128 ECB online calculator to validate with example Data
Source: AES Encryption – Easily encrypt or decrypt strings or files (online-domain-tools.com)

Seed Value: 0x 02 02 02 02 03 03 03 03 04 04 04 04 05 05 05 05 - 128bits
PrvKey : 0x 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 00 - 128bits

Encrypted Key: 0x 2B DE B7 E9 13 B1 B2 CB 95 B7 12 22 85 0B F8 76 – 128bits

Example AES ECB Encryption:



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CAN Log till Seed Value response from Server

```
;   Generated by PCAN-View v4.2.0.514
;
;   Message Number
;   |         Time Offset (ms)
;   |         |        Type
;   |         |        |       ID (hex)
;   |         |        |       |     Data Length
;   |         |        |       |     |   Data Bytes (hex) ...
;   |         |        |       |     |   |
;---+--   ----+----  --+--  ----+---  +  -+ -- -- -- -- -- -- --
    2)    44949.9  Rx        000B  8  01 00 01 0A 00 28 12 00
    3)    53399.6  Rx        07B3  8  02 3E 00 00 00 00 00 00
    4)    53419.4  Rx        071A  8  02 7E 00 BB BB BB BB BB
    5)    53438.4  Rx        07B3  8  03 22 F1 90 00 00 00 00
    6)    53449.4  Rx        071A  8  10 14 62 F1 90 54 56 53
    7)    53449.7  Rx        07B3  8  30 00 00 00 00 00 00 00
    8)    53459.5  Rx        071A  8  21 5F 47 57 4D 5F 55 33
    9)    53479.5  Rx        071A  8  22 34 37 5F 56 30 30 31
   10)    60168.3  Rx        07B3  8  02 10 02 00 00 00 00 00
   11)    60179.4  Rx        071A  8  06 50 02 00 96 00 C8 BB
   12)    64744.6  Rx        07B3  8  02 27 01 00 00 00 00 00    //SEED REQUEST FROM TESTER 27 01
   13)    64759.5  Rx        071A  8  10 0A 67 01 02 02 02 02    //FIRST FRAME SEED RESPONSE FROM
                                                                      SERVER 0X0A->10 BYTE
                                                                 RESPONSE INCLUDING SID (2 BYTE SID +
                                                                  2BYTE SEED)
   14)    65422.2  Tx        07B3  8  30 00 00 00 00 00 00 00    //FLOW CONTROL 30 00 00 REQUIRED FROM
                                                                    TESTER TO SEND REMAINING BYTES
   15)    65429.5  Rx        071A  8  21 03 03 03 03 00 00 00    //SECOND CONSECUTIVE FRAME OF SEED
                                                                     RESPONSE
   16)    69255.8  Rx        07B3  8  06 27 02 xx xx xx xx xx      // KEY RESPONSE FROM TESTER TO BE
                                                                     (KEY = 64Bits)
   17)    69255.8  Rx        071A  8  30 00 00 00 00 00 00 00
   18)    69255.8  Rx        07B3  8  21 xx xx xx 00 00 00 00
   18)    69269.5  Rx        071A  8  02 67 02 BB BB BB BB BB    // SERVER POSITIVE RESPONSE UPON
UNLOCK
                                                                      OF SECURITY LEVEL
```